Supplementary A. Results and Comparison with Baselines

This supplementary analysis details the performance of CRAT in comparison with the baselines.

A.1 Evaluation scope and overall performance

We compare CRAT with the unadopted backbone, Full SFT, Targeted SFT, LoRA, rsLoRA, and DoRA on the EN-FR, EN-FA, and EN-ZH language pairs from OPUS100, using Llama 2 7B and Mistral 7B. Each reported quality score is the mean across six seeds. Tables 2 and 3 report these means with the accompanying standard deviations (SDs). BLEU and chrF retain their reported point scales, while COMET retains its original decimal scale. Higher values indicate better performance. Pair-level scores are analyzed as reported; they do not identify the performance of either translation direction separately. BLEU, chrF, and COMET provide complementary automatic evaluation perspectives.

CRAT ranks first in nine of the 18 model-pair-metric comparisons and second in the remaining nine. It achieves three first-place means on each metric. Across the six model-pair settings, CRAT exceeds the base model and Targeted SFT in all 18 metric comparisons, Full SFT and rsLoRA in 15, LoRA in 16, and DoRA in 17. These counts describe rankings of reported means; the correlated metrics are not independent replications, and a higher mean alone does not establish statistical significance.

Table 1 summarizes performance using an unweighted arithmetic mean over the six model-pair settings, computed separately for each metric. CRAT has the highest macro-average BLEU, chrF, and COMET and the lowest average rank. Its closest aggregate competitor is rsLoRA: CRAT improves the macro averages by 0.169 BLEU, 0.140 chrF, and 0.00160 COMET. The corresponding gains over Targeted SFT are 0.944 BLEU, 0.499 chrF, and 0.00647 COMET. These summaries describe this benchmark suite; they are not pooled corpus scores or estimates of performance on an unseen language population.

Table 1. Unweighted macro averages over the six model-pair settings and descriptive ranks.

Method	BLEU	chrF	COMET	Mean rank	First places
Base	15.570	35.980	0.70086	6.72	0
Full SFT	18.319	38.320	0.72732	5.11	3
Targeted SFT	19.978	40.194	0.74638	4.22	0
LoRA	20.419	39.906	0.74430	3.78	2
rsLoRA	20.753	40.554	0.75125	2.67	3
DoRA	20.234	40.030	0.74486	4.00	1
CRAT	**20.922**	**40.693**	**0.75285**	**1.50**	**9**

Higher is better for quality and first-place count; lower is better for mean rank. Rank averages cover 18 metric-setting cells. Macro scores are arithmetic averages of reported means, not pooled corpus scores. Bold marks the best value.

A.2 Complete quality results across language pairs

English and French. On Llama 2, CRAT obtains 29.295 BLEU, 53.874 chrF, and 0.80297 COMET. Its BLEU is effectively tied at the displayed precision with LoRA: the unrounded CRAT-minus-LoRA difference is -0.000464. CRAT has the highest chrF and COMET means, exceeding the strongest respective baselines by 0.199 and 0.00265. On Mistral, CRAT reaches 31.358 BLEU, 55.763 chrF, and 0.80679 COMET. It leads BLEU by 0.023 over DoRA, while DoRA leads chrF by 0.00569 and LoRA leads COMET by only 0.0000353. The small differences among these leading methods support a conclusion of closely matched observed performance, without establishing statistical equivalence. Full SFT is substantially weaker on Mistral EN-FR, with 24.056 BLEU, 49.124 chrF, and 0.74770

COMET; all three means are below the unadopted Mistral baseline. This result concerns the evaluated configuration and does not establish that full fine-tuning is intrinsically inferior.

Table 2. Llama 2 7B translation quality on OPUS100. Values are mean ± reported SD over six seeds.

Method	BLEU	chrF	COMET
EN-FR			
Base	26.493 ± 0.051	52.994 ± 0.019	0.79026 ± 0.00001
Full SFT	27.353 ± 0.091	52.032 ± 0.191	0.78880 ± 0.00246
Targeted SFT	28.729 ± 0.282	53.548 ± 0.200	0.79914 ± 0.00055
LoRA	**29.295 ± 0.357**	53.675 ± 0.323	0.80032 ± 0.00150
rsLoRA	29.118 ± 0.047	53.676 ± 0.083	0.79994 ± 0.00122
DoRA	29.291 ± 0.295	53.641 ± 0.336	0.80019 ± 0.00153
CRAT †	29.295 ± 0.089	**53.874 ± 0.104**	**0.80297 ± 0.00097**
EN-FA			
Base	3.457 ± 0.002	18.717 ± 0.006	0.52261 ± 0.00029
Full SFT	**6.535 ± 0.062**	**23.846 ± 0.222**	**0.63719 ± 0.00290**
Targeted SFT	5.680 ± 0.138	22.805 ± 0.261	0.61912 ± 0.00524
LoRA	5.440 ± 0.035	21.517 ± 0.058	0.59592 ± 0.00050
rsLoRA	6.013 ± 0.054	23.251 ± 0.092	0.62873 ± 0.00176
DoRA	5.677 ± 0.041	22.021 ± 0.072	0.60038 ± 0.00042
CRAT	6.402 ± 0.094	23.554 ± 0.124	0.62949 ± 0.00506
EN-ZH			
Base	15.817 ± 0.018	35.040 ± 0.053	0.76816 ± 0.00023
Full SFT	22.660 ± 0.542	39.860 ± 0.107	0.78335 ± 0.00397
Targeted SFT	21.983 ± 0.639	39.833 ± 0.203	0.79269 ± 0.00099
LoRA	22.017 ± 0.043	39.201 ± 0.121	0.79069 ± 0.00119
rsLoRA	22.713 ± 0.263	**40.521 ± 0.177**	0.79509 ± 0.00061
DoRA	21.634 ± 0.287	39.285 ± 0.135	0.79111 ± 0.00064
CRAT	**22.807 ± 0.209**	40.047 ± 0.013	**0.79769 ± 0.00201**

Higher is better. Bold marks the highest unrounded mean within each pair and metric; shaded rows identify language-pair. Rounded scores can appear tied even when the unrounded ranking differs. † Llama 2 EN-FR CRAT used 158 updates and 1,582,308 tokens; the adapted baselines used 316 updates and 3,164,616 tokens.

English and Persian. Llama 2 EN-FA is the clearest exception to CRAT's leading quality profile. Full SFT achieves the highest means on all three metrics: 6.535 BLEU, 23.846 chrF, and 0.63719 COMET. CRAT ranks second, with deficits of 0.134 BLEU, 0.293 chrF, and 0.00770 COMET. It nevertheless exceeds every parameter-efficient baseline on all three metrics. On Mistral EN-FA, CRAT leads BLEU and chrF with 8.685 and 27.176, respectively. Its advantages over the strongest baselines are 0.020 BLEU over rsLoRA and 0.083 chrF over Targeted SFT. rsLoRA

has the highest COMET mean, exceeding CRAT by 0.00104. Thus, the relative strength of Full SFT versus selective or low-rank adaptation changes substantially between the two backbones for this pair.

English and Chinese. On Llama 2 EN-ZH, CRAT achieves the highest BLEU and COMET means, exceeding rsLoRA by 0.094 BLEU and 0.00260 COMET. However, rsLoRA has a 0.474-point chrF advantage, the largest chrF deficit of CRAT against a baseline in this comparison. On Mistral EN-ZH, rsLoRA leads BLEU by 0.018, while CRAT leads chrF by 0.065 over rsLoRA and COMET by 0.00251 over Targeted SFT. CRAT improves on Targeted SFT by 2.538 BLEU in this setting, its largest BLEU gain over that baseline. The disagreement between metric rankings cautions against attributing a uniform linguistic improvement to the aggregate scores. In particular, higher COMET is evidence from a learned evaluator, not a direct measurement of morphological correctness or human translation adequacy.

Table 3. Mistral 7B translation quality on OPUS100. Values are mean ± reported SD over six seeds.

Method	BLEU	chrF	COMET
EN-FR			
Base	27.072 ± 0.052	52.233 ± 0.019	0.78313 ± 0.00001
Full SFT	24.056 ± 0.080	49.124 ± 0.180	0.74770 ± 0.00233
Targeted SFT	30.592 ± 0.301	54.793 ± 0.204	0.80443 ± 0.00055
LoRA	31.209 ± 0.380	55.557 ± 0.334	**0.80683 ± 0.00151**
rsLoRA	31.006 ± 0.050	55.200 ± 0.086	0.80576 ± 0.00123
DoRA	31.335 ± 0.316	**55.768 ± 0.349**	0.80671 ± 0.00154
CRAT	**31.358 ± 0.096**	55.763 ± 0.108	0.80679 ± 0.00098
EN-FA			
Base	5.243 ± 0.003	21.937 ± 0.007	0.58414 ± 0.00032
Full SFT	6.401 ± 0.061	23.695 ± 0.220	0.63375 ± 0.00288
Targeted SFT	8.435 ± 0.205	27.093 ± 0.310	0.65785 ± 0.00557
LoRA	8.361 ± 0.054	26.232 ± 0.070	0.66784 ± 0.00056
rsLoRA	8.665 ± 0.078	26.992 ± 0.107	**0.67363 ± 0.00188**
DoRA	8.256 ± 0.059	26.326 ± 0.086	0.66698 ± 0.00046
CRAT	**8.685 ± 0.127**	**27.176 ± 0.143**	0.67259 ± 0.00541
EN-ZH			
Base	15.340 ± 0.017	34.959 ± 0.053	0.75687 ± 0.00023
Full SFT	22.910 ± 0.548	41.365 ± 0.111	0.77311 ± 0.00392
Targeted SFT	24.446 ± 0.711	43.093 ± 0.220	0.80506 ± 0.00101
LoRA	26.191 ± 0.051	43.256 ± 0.134	0.80419 ± 0.00121
rsLoRA	**27.001 ± 0.313**	43.681 ± 0.191	0.80437 ± 0.00061
DoRA	25.214 ± 0.335	43.137 ± 0.148	0.80382 ± 0.00065
CRAT	26.984 ± 0.247	**43.746 ± 0.014**	**0.80757 ± 0.00203**

Higher is better. Bold marks the highest unrounded mean within each pair and metric; shaded rows identify language-pair. Rounded scores can appear tied even when the unrounded ranking differs.

A.3 Consistency across metrics and backbones

Figure 1 compares CRAT with the strongest baseline selected independently for each metric and setting. This stricter comparison exposes both its advantages and its exceptions. CRAT never falls below second place, but no setting yields an outright first-place mean on all three metrics. Several apparent rank differences are numerically negligible, particularly BLEU on Llama 2 EN-FR, COMET on Mistral EN-FR, and BLEU on Mistral EN-FA and EN-ZH. Rankings must therefore be read together with the absolute margins and reported variability.

The Mistral CRAT configuration has higher means than the Llama 2 CRAT configuration on all nine pair-metric comparisons. Its BLEU advantages are 2.063 on EN-FR, 2.283 on EN-FA, and 4.177 on EN-ZH; its COMET advantages are 0.00382, 0.04310, and 0.00987, respectively. These differences describe the resulting systems and do not isolate an architectural effect. The unadopted backbones and baselines differ as well, and Llama 2 EN-FR also has an unequal CRAT training budget. Within each backbone, CRAT improves over the base model on every metric; the BLEU gains range from 2.802 to 11.644 points.

A.4 Variation across seeds

CRAT's variability depends on the language pair and metric. Its reported BLEU SD ranges from 0.089 to 0.247, and its COMET SD ranges from 0.00097 to 0.00541. The EN-FA COMET SDs are 0.00506 for Llama 2 and 0.00541 for Mistral, compared with 0.00176 and 0.00188 for rsLoRA. These values do not support a general claim that CRAT reduces seed sensitivity. Conversely, its EN-ZH chrF SDs are only 0.0127 and 0.0138, the smallest among the adapted methods for that metric in the respective settings. High mean performance and low variability should therefore be treated as separate properties.

A.5 Trainable parameters and memory requirements

CRAT updates 195.53-256.35 million parameters, corresponding to 2.63-3.66% of the registered model parameters in these configurations. Relative to the trainable parameter count of Full SFT, this is a reduction of 96.20-97.30%. The registered-model percentage and the reduction relative to Full SFT use different denominators: selective adaptation

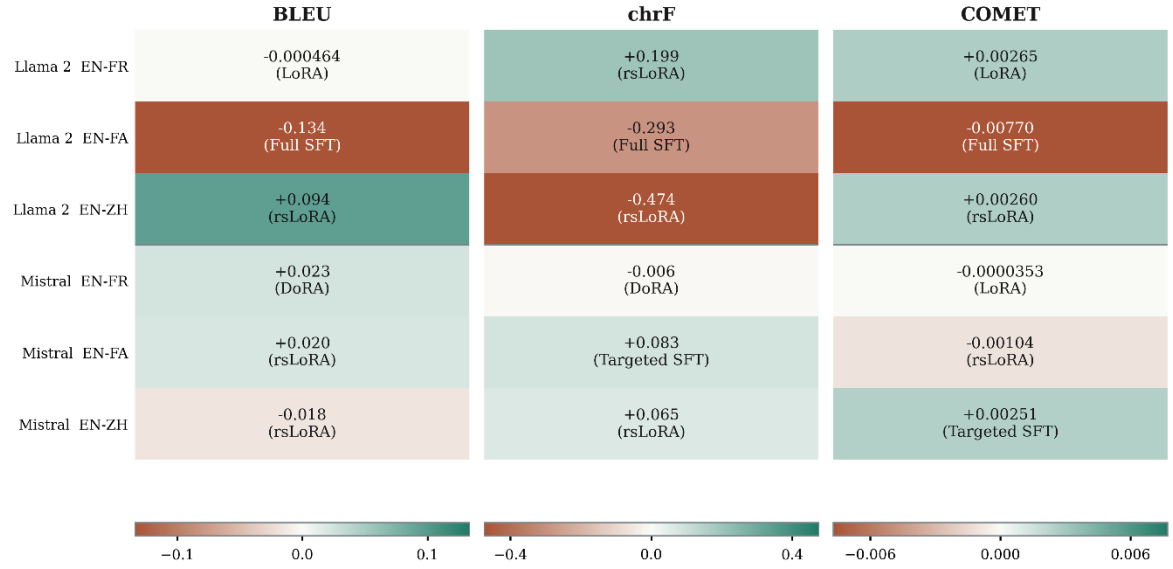


Figure 1. CRAT minus the strongest non-CRAT mean in each metric and setting. Comparator selection is metric specific. Values are absolute score differences computed before rounding; no significance test is implied.

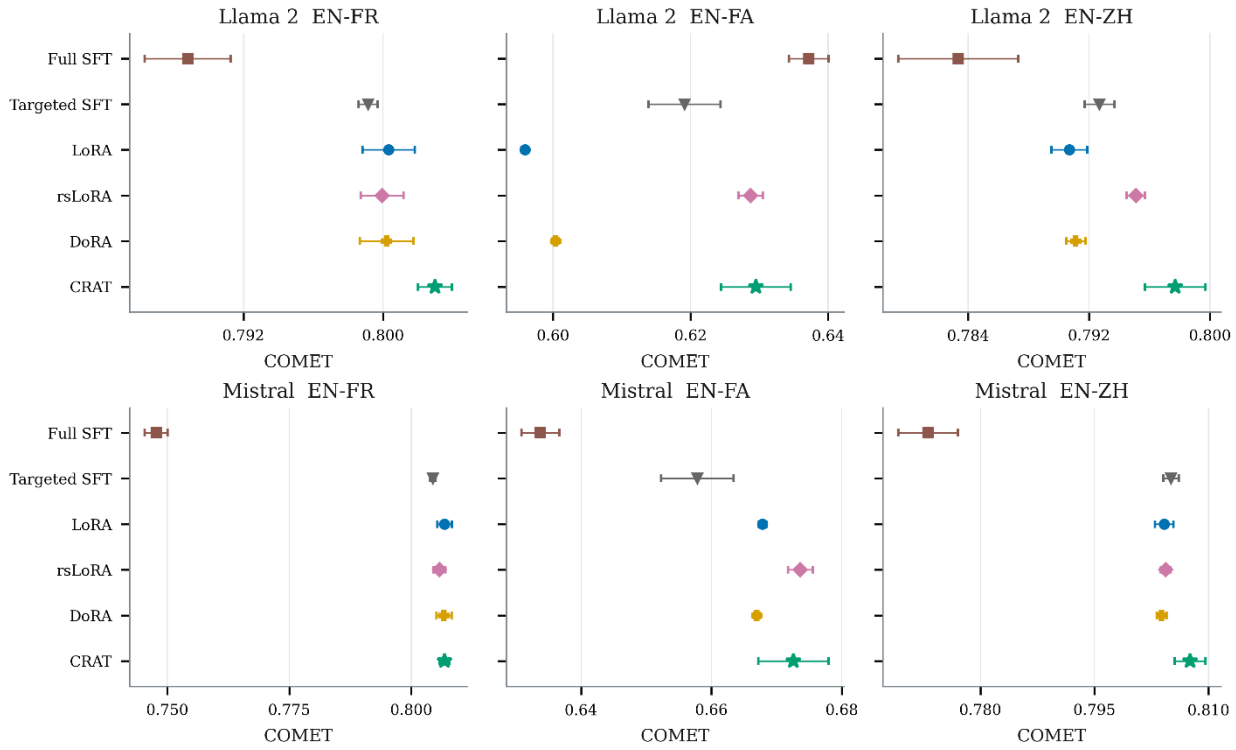


Figure 2. COMET performance of all adapted methods. Points are six-seed means and bars show ±1 reported SD. The horizontal scale differs across panels. Error bars do not represent uncertainty in a paired method difference.

retains the frozen backbone and registers additional trainable parameters. CRAT adds exactly 4,688,685 trainable parameters to Targeted SFT in every setting, an increase of 1.86-2.46%. Together with its improvements in all 18 quality comparisons, this supports a favorable quality gain for a modest parameter increment over that particular baseline.

LoRA and rsLoRA remain substantially smaller, using 39.98 million trainable parameters on Llama 2 and 41.94 million on Mistral. CRAT uses 4.66-6.41 times as many trainable parameters. Its peak VRAM values are 37.36-41.24 in the reported field, only 0.06-4.10% below Full SFT and 57.49-66.71% above LoRA or rsLoRA. CRAT also uses 60.19-64.05% more peak VRAM than Targeted SFT. Thus, parameter savings do not translate into proportional peak-memory savings. The CRAT loads a frozen teacher when KL supervision is enabled, which can contribute to this overhead; the aggregate measurements do not isolate its individual cost. Tables S1 and S2 give the full parameter and training results, including the precise memory-unit convention.

A.6 Training and generation costs

The reported CRAT training duration is 18.67-29.74 minutes, compared with 64.00-79.89 minutes for Full SFT. CRAT reduces the reported training time by 62.52-76.26%; excluding the unequal-budget Llama 2 EN-FR setting, the range is 62.52-71.50%. This advantage does not extend uniformly to the smaller adaptation baselines. In the other five settings, CRAT takes 37.29-51.47% longer than Targeted SFT and 0.62-11.00% longer than LoRA. Compared with rsLoRA, it is 0.45% faster on Mistral EN-FR but 7.08-10.45% slower in the four remaining budget-comparable settings. Relative to DoRA, CRAT uses 33.70-36.80% less peak VRAM and 53.65-68.20% less training time across the five budget-comparable settings, despite its larger trainable parameter count.

Training budget and measurement scope. Llama 2 EN-FR requires separate interpretation. CRAT records 158 updates and 1,582,308 training tokens, whereas every adapted baseline records 316 updates and 3,164,616 tokens. Its 1,127.50-second duration, compared with 1,165.76 seconds for Targeted SFT, therefore does not demonstrate a lower cost for the same workload. CRAT's reported training throughput is 1,403.38 tokens/s, versus 2,714.64 for Targeted

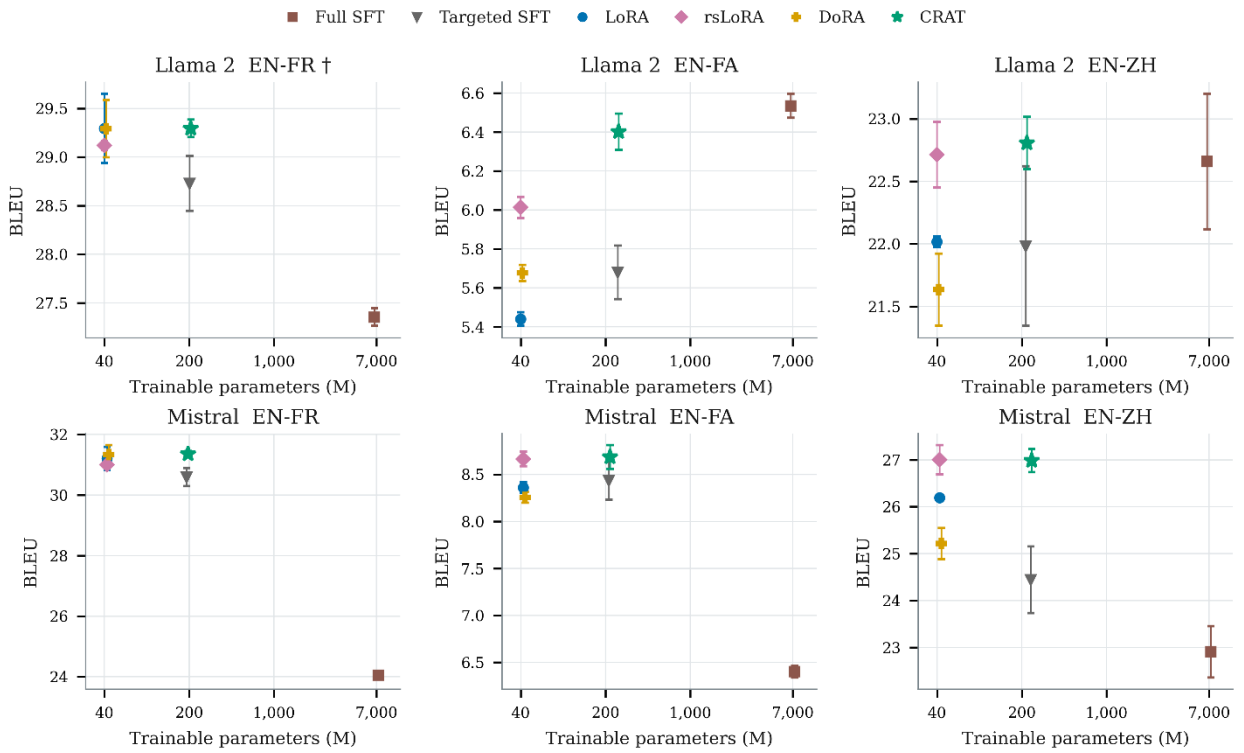


Figure 3. Observed BLEU versus trainable parameter count. The horizontal axis is logarithmic and vertical bars show ±1 reported SD. Zero-trainable-parameter base models are omitted. The dagger identifies the unequal training budget on Llama 2 EN-FR.

SFT and 1,995.05 for LoRA. The quality obtained under this shorter schedule is encouraging, but establishing better sample efficiency requires matched learning curves or a controlled budget comparison. The other settings have matching update counts within each comparison; very small token-count differences between Full SFT and the other methods remain in several files.

The timing boundary also matters. In the supplied implementation, component discovery and morphology preprocessing occur before the training timer; the reported training interval includes training-loop work and validation. Consequently, the recorded training time does not represent the entire adaptation pipeline. Likewise, the generation timer starts after prompt tokenization, source-feature construction, and CRAT router execution. The latency results should be described as timed generation latency, with the precise execution revision confirmed for the reported experiments.

Generation throughput and latency. CRAT's batched generation throughput is 50.70-102.92 tokens/s. It exceeds LoRA and rsLoRA in five of six settings, but Mistral EN-ZH reverses this pattern: CRAT reaches 102.92 tokens/s versus 110.80 for LoRA and 109.84 for rsLoRA, deficits of 7.11% and 6.30%. Targeted SFT has higher batched throughput in every setting, and Full SFT has higher throughput in five. The largest CRAT throughput advantage is over the evaluated DoRA implementation, ranging from 2.98 to 5.40 times. This is an implementation- and configuration-specific result, not a universal property of either method.

The single-example benchmark gives a similarly conditional picture. CRAT's mean timed generation latency is 0.954-2.814 seconds. It is higher than Full SFT and Targeted SFT in all six settings, by 24.22-113.49% and 8.20-49.50%, respectively. It is lower than LoRA in four settings and rsLoRA in five, and lower than DoRA in all six. Tail behavior is not identical to mean behavior: CRAT has lower p95 latency than LoRA in every setting, but higher p95 latency than rsLoRA on Llama 2 EN-FA and EN-ZH. Tables S3 and S4 report the mean, median, and p95 latencies separately.

Raw generation duration must also be interpreted with output length. The base models generate approximately 123,909-127,284 tokens, compared with 27,167-36,578 for CRAT, a 3.48-4.56-fold difference within matched

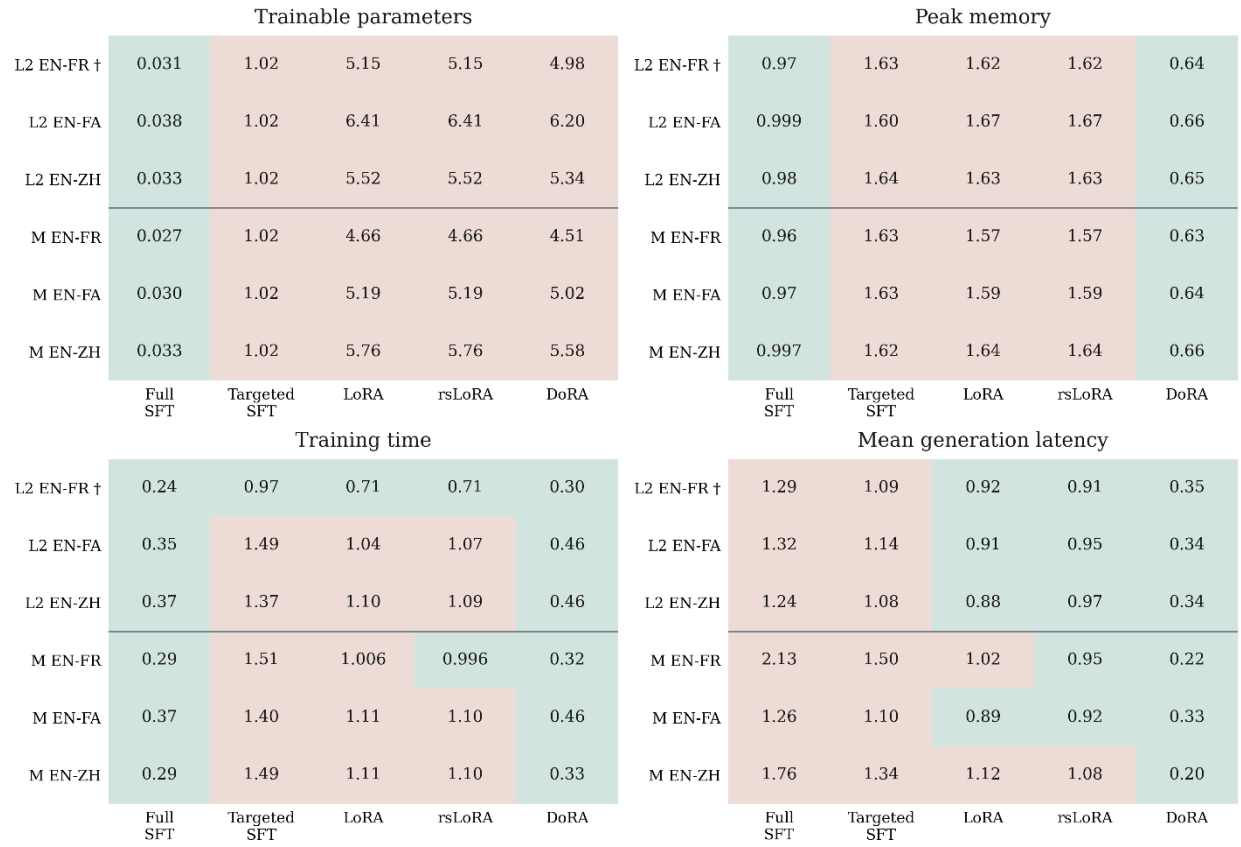


Figure 4. CRAT-to-baseline ratios of reported means. Lower values favor CRAT for all four costs. Colors encode direction, not statistical significance. Training time in the dagger-marked setting reflects half as many CRAT updates and tokens.

settings. Their higher tokens/s does not by itself indicate faster delivery of useful translations. The summaries do not reveal whether longer outputs reflect verbosity, repetition, stopping behavior, or another cause; that question requires inspecting the generated text. Batched throughput and single-example benchmark throughput are distinct workloads and should not be substituted for one another.

Supplementary Comparison Tables

Table S1. Llama 2 7B trainable parameters, peak memory, and reported training cost.

Method	Trainable (M)	Trainable (%)	Peak VRAM*	Train (min)	Train (tokens/s)	Updates
EN-FR						
Base	0.00	0.00	13.96	0.00	0.0	0
Full SFT	6,738.42	100.00	38.54	79.15	666.4	316
Targeted SFT	201.33	2.90	22.91	19.43	2,714.6	316
LoRA	39.98	0.59	23.10	26.44	1,995.0	316
rsLoRA	39.98	0.59	23.10	26.53	1,988.2	316
DoRA	41.34	0.61	58.10	62.21	847.8	316
CRAT †	206.02	2.97	37.36	18.79	1,403.4	158
EN-FA						
Base	0.00	0.00	14.10	0.00	0.0	0
Full SFT	6,738.42	100.00	38.54	78.75	640.2	312
Targeted SFT	251.66	3.60	24.04	18.54	2,718.7	312

LoRA	39.98	0.59	23.10	26.44	1,906.3	312
rsLoRA	39.98	0.59	23.10	25.80	1,953.9	312
DoRA	41.34	0.61	58.10	59.60	845.8	312
CRAT	256.35	3.66	38.52	27.63	1,824.8	312
EN-ZH						
Base	0.00	0.00	14.10	0.00	0.0	0
Full SFT	6,738.42	100.00	38.54	79.89	780.4	312
Targeted SFT	216.01	3.11	22.91	21.66	2,877.9	312
LoRA	39.98	0.59	23.10	27.09	2,301.6	312
rsLoRA	39.98	0.59	23.10	27.30	2,283.5	312
DoRA	41.34	0.61	58.10	64.81	962.0	312
CRAT	220.70	3.17	37.59	29.74	2,096.1	312

M denotes millions of parameters. Trainable percentage uses each method's registered total parameter count. Resource entries are reported means; available resource SDs are retained in the companion CSV. Base training entries are zero because no adaptation is performed.

† EN-FR CRAT has half the updates and training tokens of every adapted baseline. Other Llama 2 pairs have 312 updates. The effective batch size field is 128 for all rows.

Table S2. Mistral 7B trainable parameters, peak memory, and reported training cost.

Method	Trainable (M)	Trainable (%)	Peak VRAM*	Train (min)	Train (tokens/s)	Updates
EN-FR						
Base	0.00	0.00	13.93	0.00	0.0	0
Full SFT	7,241.73	100.00	41.37	64.00	819.0	316
Targeted SFT	190.84	2.57	24.34	12.33	4,247.6	316
LoRA	41.94	0.58	25.19	18.56	2,822.8	316
rsLoRA	41.94	0.58	25.19	18.76	2,791.7	316
DoRA	43.32	0.59	62.76	58.71	891.8	316
CRAT	195.53	2.63	39.67	18.67	2,804.7	316
EN-FA						
Base	0.00	0.00	14.00	0.00	0.0	0
Full SFT	7,241.73	100.00	41.37	77.93	628.4	312
Targeted SFT	212.86	2.86	24.56	20.90	2,343.1	312
LoRA	41.94	0.58	25.19	26.32	1,860.4	312
rsLoRA	41.94	0.58	25.19	26.44	1,851.8	312
DoRA	43.32	0.59	62.76	63.17	775.1	312
CRAT	217.55	2.92	40.02	29.20	1,676.6	312
EN-ZH						
Base	0.00	0.00	14.00	0.00	0.0	0
Full SFT	7,241.73	100.00	41.37	69.24	808.7	316
Targeted SFT	236.98	3.17	25.51	13.20	4,241.2	316
LoRA	41.94	0.58	25.19	17.78	3,149.4	316
rsLoRA	41.94	0.58	25.19	17.94	3,121.0	316
DoRA	43.32	0.59	62.76	60.38	927.4	316
CRAT	241.67	3.23	41.24	19.74	2,837.2	316

M denotes millions of parameters. Trainable percentage uses each method's registered total parameter count. Resource entries are reported means; available resource SDs are retained in the companion CSV. Base training entries are zero because no adaptation is performed.

All adapted EN-FR and EN-ZH rows have 316 updates; EN-FA rows have 312. The effective batch size field is 128 for all rows. The Mistral EN-FR export additionally reports resource SDs.

Table S3. Llama 2 7B generation duration, output volume, throughput, and latency.

Method	Gen. (s)	Generated tokens	Batch tokens/s	Mean (s)	p50 (s)	p95 (s)	Single tokens/s
EN-FR							
Base	592.8	123,909	209.0	2.342	1.856	3.739	31.6
Full SFT	478.3	26,513	55.4	1.319	0.622	2.597	18.5
Targeted SFT	452.2	26,833	59.3	1.558	0.687	2.955	15.6
LoRA	548.9	26,691	48.6	1.852	0.936	3.431	12.9
rsLoRA	552.1	26,716	48.4	1.856	0.964	3.394	13.0
DoRA	1,528.7	26,650	17.4	4.853	2.816	12.639	4.9
CRAT	494.6	27,167	54.9	1.695	0.849	3.328	14.6
EN-FA							
Base	644.2	126,715	196.7	2.837	2.628	5.093	41.7
Full SFT	493.4	28,016	56.8	1.439	0.696	5.430	18.2
Targeted SFT	548.2	29,595	54.0	1.668	0.781	8.105	15.8
LoRA	680.9	30,745	45.2	2.082	1.001	9.558	13.0
rsLoRA	659.7	29,019	44.0	1.996	0.986	7.157	12.8
DoRA	1,790.0	30,469	17.0	5.548	2.906	18.170	4.9
CRAT	626.3	31,751	50.7	1.893	0.913	7.663	14.8
EN-ZH							
Base	606.2	127,284	210.0	3.526	2.658	10.687	35.2
Full SFT	583.4	36,618	62.8	2.265	0.926	7.780	17.1
Targeted SFT	523.2	36,565	69.9	2.601	1.230	6.708	15.9
LoRA	662.7	36,573	55.2	3.201	1.667	8.047	12.9
rsLoRA	666.4	36,655	55.0	2.911	1.665	5.406	13.9
DoRA	1,930.1	36,623	19.0	8.237	5.410	20.141	5.0
CRAT	566.5	36,578	64.6	2.814	1.412	6.381	14.8

Gen. is total timed generation duration over the evaluation workload. Batch tokens/s is generation tokens per second. Mean, p50, p95, and Single tokens/s come from the separate single-example latency benchmark. They are not alternative summaries of the batched workload. All entries are the reported means; p50 and p95 are means of run-level percentile statistics when aggregated across seeds.

The batch generate timer starts after tokenization, source features, and CRAT router computation. These timings exclude that setup work. Different output token counts also affect raw duration; tokenizer differences limit cross-backbone interpretation of tokens/s.

Table S4. Mistral 7B generation duration, output volume, throughput, and latency.

Method	Gen. (s)	Generated tokens	Batch tokens/s	Mean (s)	p50 (s)	p95 (s)	Single tokens/s
EN-FR							
Base	272.1	124,521.5	457.8	1.660	1.673	2.073	72.6
Full SFT	219.2	27,356.5	125.0	0.447	0.273	1.271	54.4
Targeted SFT	205.9	26,951	130.9	0.638	0.350	1.824	37.8
LoRA	300.2	27,200.5	91.2	0.936	0.494	2.608	26.0
rsLoRA	311.1	27,219	87.5	0.999	0.509	2.547	24.1
DoRA	1,484.3	27,280	18.4	4.430	2.608	14.379	5.4
CRAT	295.9	27,298.5	92.9	0.954	0.506	2.447	26.1
EN-FA							
Base	677.7	126,087	186.0	3.230	2.630	9.682	38.9
Full SFT	468.5	27,246	58.2	1.475	0.678	6.379	18.6
Targeted SFT	534.0	28,243	52.9	1.679	0.777	4.934	15.0
LoRA	615.1	28,013	45.5	2.078	0.995	10.541	12.8
rsLoRA	607.0	28,022	46.2	2.021	0.944	8.576	12.7

DoRA	1,715.8	27,757	16.2	5.659	3.138	19.949	4.5
CRAT	563.6	29,226	51.9	1.854	0.855	7.604	13.8
EN-ZH							
Base	265.3	124,090	467.8	1.732	1.734	2.080	69.8
Full SFT	257.5	32,807	127.4	0.732	0.433	1.576	51.1
Targeted SFT	236.9	31,505	133.0	0.963	0.611	2.553	37.4
LoRA	286.1	31,705	110.8	1.149	0.797	3.237	31.1
rsLoRA	292.4	32,112	109.8	1.192	0.743	3.303	30.6
DoRA	1,662.6	31,676	19.1	6.481	4.728	19.564	5.4
CRAT	309.0	31,804	102.9	1.292	0.810	3.195	27.5

Gen. is total timed generation duration over the evaluation workload. Batch tokens/s is generation tokens per second. Mean, p50, p95, and Single tokens/s come from the separate single-example latency benchmark. They are not alternative summaries of the batched workload. All entries are the reported means; p50 and p95 are means of run-level percentile statistics when aggregated across seeds.

The batch generate timer starts after tokenization, source features, and CRAT router computation. These timings exclude that setup work. Different output token counts also affect raw duration; tokenizer differences limit cross-backbone interpretation of tokens/s.